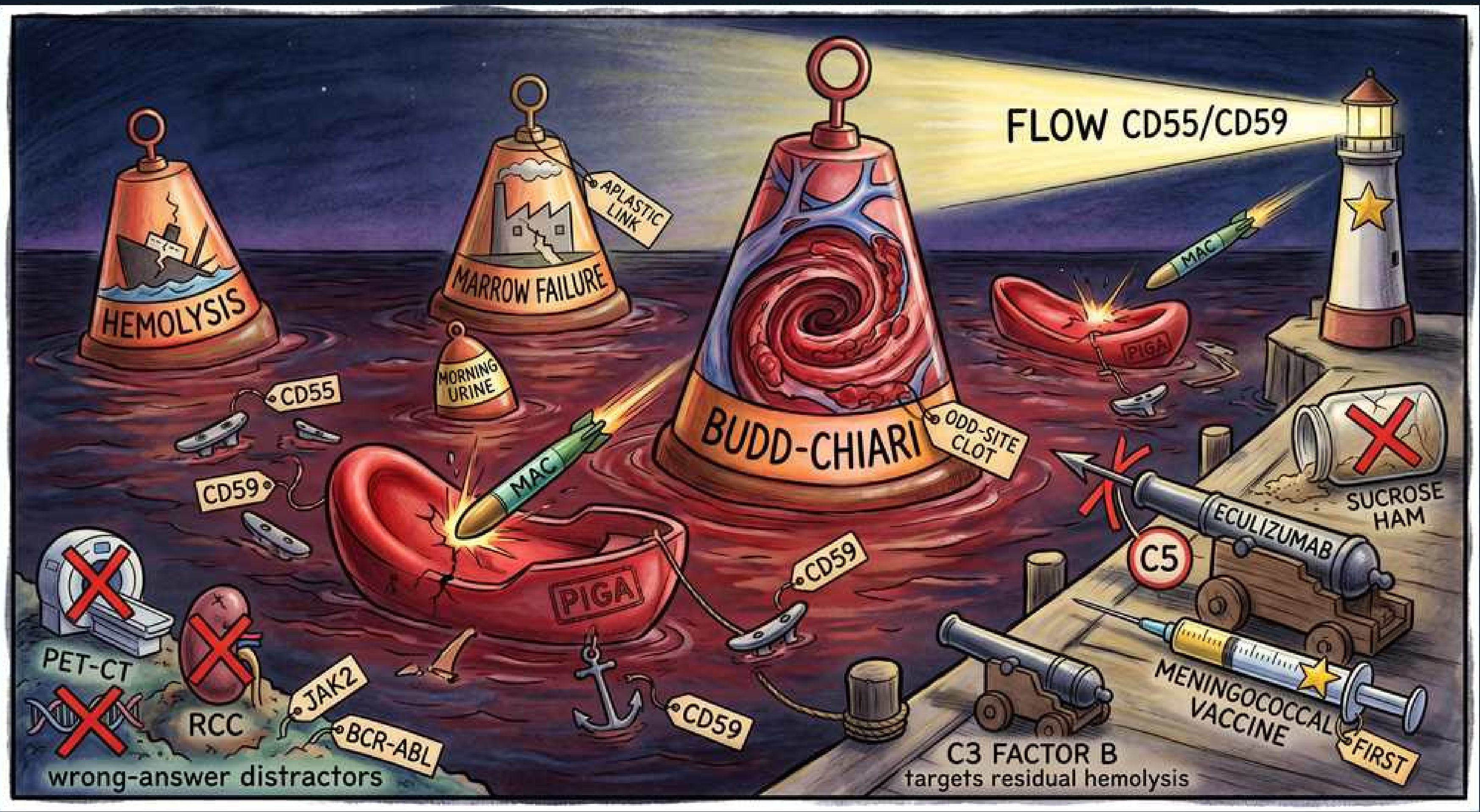
Paroxysmal Nocturnal Hemoglobinuria (PNH)



1. PIGA mutation

WHAT YOU SEE
The 'PIGA' banner stamped across the hull of the foundering red-cell ship — the acquired keel-flaw that crippled the vessel.

WHAT IT ENCODES
PNH is caused by an ACQUIRED somatic mutation in PIGA (X-linked) within a hematopoietic stem cell, abolishing synthesis of the GPI anchor.

BOARD TRAP
Calling it inherited/germline — PNH is an acquired clonal stem-cell disorder.

2. CD55 / CD59 lost

WHAT YOU SEE
The broken-off anchor-cleat tags reading 'CD55' and 'CD59' tumbling off the deck, leaving the ship without its complement armor.

WHAT IT ENCODES
The GPI anchor normally tethers the complement regulators CD55 (DAF) and CD59 (MIRL); their loss leaves red cells defenseless against complement, with CD59 loss the main driver of hemolysis.

3. Complement / MAC

WHAT YOU SEE
The lit complement torpedo labeled 'MAC' streaking into and cracking the red-cell ship's hull.

WHAT IT ENCODES
Unregulated complement forms the membrane-attack complex, producing chronic intravascular hemolysis.

BOARD TRAP
Expecting extravascular/splenic hemolysis — PNH is complement-mediated intravascular.

4. Morning urine (hemoglobinuria)

WHAT YOU SEE
The small buoy tagged 'MORNING URINE' bobbing in the dark night water beside the ships.

WHAT IT ENCODES
"Paroxysmal nocturnal" — hemoglobinuria is classically worst in concentrated morning urine; chronic urinary iron loss causes iron deficiency.

BOARD TRAP
Calling it gross hematuria — it is hemoglobinuria (dipstick-positive, no red cells).

5. Hemolysis (triad)

WHAT YOU SEE
The far-left 'HEMOLYSIS' triad buoy, a cone marking a sinking ship in the water.

WHAT IT ENCODES
Part of the triad: a Coombs-NEGATIVE intravascular hemolysis with high LDH and low haptoglobin.

BOARD TRAP
Expecting a positive DAT — PNH hemolysis is Coombs-negative.

6. Marrow failure (triad)

WHAT YOU SEE
The 'MARROW FAILURE' triad buoy showing a crumbling factory, its 'ASPLASTIC LINK' tag tying it to aplastic anemia.

WHAT IT ENCODES
The triad also includes cytopenias / bone-marrow failure; PNH clones arise from and coexist with aplastic anemia and MDS.

7. Budd-Chiari — odd-site thrombosis

WHAT YOU SEE
The towering central 'BUDD-CHIARI' buoy whose red whirlpool churns like a hepatic-vein clot, the 'ODD-SITE CLOT' tag pinned to its side.

WHAT IT ENCODES
THROMBOSIS is the leading cause of death in PNH, classically at unusual sites — the hepatic vein (Budd-Chiari), other intra-abdominal veins, and cerebral veins. Unexplained thrombosis plus hemolysis or cytopenia should trigger a PNH work-up.

BOARD TRAP
Missing PNH in a Budd-Chiari + hemolysis vignette — odd-site venous thrombosis with hemolysis is the classic PNH signal.

8. Flow cytometry — CD55/CD59

WHAT YOU SEE
The beaming 'FLOW CD55/CD59' lighthouse on the far-right pier, sweeping its light to scan ships for missing anchor-proteins.

WHAT IT ENCODES
Diagnosis is by FLOW CYTOMETRY showing loss of GPI-anchored proteins (CD55/CD59) on red cells and granulocytes; FLAER is the most sensitive assay.

BOARD TRAP
Choosing osmotic fragility, sucrose or Ham as the diagnostic test — flow cytometry is the modern gold standard.

9. Sucrose / Ham (obsolete)

WHAT YOU SEE
The red-X-crossed 'SUCROSE HAM' bottle of sugar-water lying discarded on the dock.

WHAT IT ENCODES
The old sugar-water (sucrose lysis) and Ham acid-hemolysis tests are obsolete, replaced by flow cytometry.

BOARD TRAP
Ordering a Ham or sucrose test as the answer — outdated.

10. Eculizumab — anti-C5

WHAT YOU SEE

The 'ECULIZUMAB' dock cannon aimed to intercept torpedoes, its 'C5' cannonball blocking terminal complement.

WHAT IT ENCODES

Eculizumab and ravulizumab are anti-C5 monoclonals that block terminal complement, controlling intravascular hemolysis and thrombosis.

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11. Meningococcal vaccine FIRST

WHAT YOU SEE

The 'MENINGOCOCCAL VACCINE' syringe stamped 'FIRST', planted on the dock to be given before the cannon may fire.

WHAT IT ENCODES

C5 blockade prevents membrane-attack-complex formation, creating a high risk of Neisseria meningitidis infection, so patients must be VACCINATED against meningococcus (at least 2 weeks before) starting eculizumab.

BOARD TRAP

Starting eculizumab without meningococcal vaccination — the tested safety step.

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12. C3 / Factor B inhibitors

WHAT YOU SEE

The pair of smaller dock cannons labeled 'C3 FACTOR B — targets residual hemolysis,' mopping up the survivors.

WHAT IT ENCODES

Residual extravascular hemolysis from C3 opsonization on anti-C5 therapy is treated with proximal inhibitors: pegcetacoplan (C3), iptacopan (factor B) and danicopan (factor D).

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13. Distractors — PET-CT / RCC / JAK2 / BCR-ABL

WHAT YOU SEE

The red-X-crossed 'wrong-answer distractors' reef at bottom-left — PET-CT, RCC, JAK2 and BCR-ABL rocks luring you off course.

WHAT IT ENCODES

Common wrong answers: PET-CT (not the work-up), renal cell carcinoma / renal vein thrombosis (a non-PNH thrombosis distractor), JAK2 (polycythemia vera / MPN), and BCR-ABL (CML).

BOARD TRAP

Attributing the thrombosis to RCC or an MPN (JAK2/BCR-ABL) instead of testing for PNH by flow cytometry.
